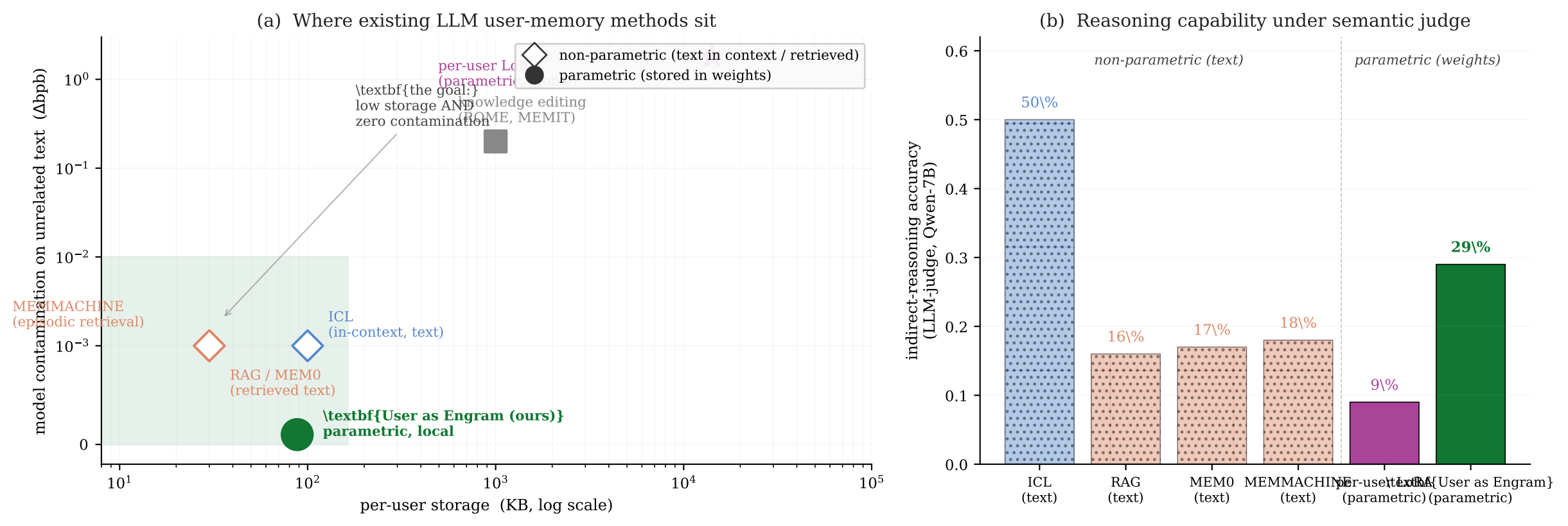

(a) Where existing LLM user-memory methods sit
non-parametric (text in context / retrieved)
parametric (stored in weights)
per-user LoRA
(parametric)
knowledge editing
(ROME, MEMIT)
\textbf{the goal:}
low storage AND
zero contamination
MEMMACHINE
(episodic retrieval)
ICL
(in-context, text)
RAG / MEM0
(retrieved text)
\textbf{User as Engram (ours)}
parametric, local
model contamination on unrelated text (Δbpb)
per-user storage (KB, log scale)
(b) Reasoning capability under semantic judge
non-parametric (text)
parametric (weights)
indirect-reasoning accuracy
(LLM-judge, Qwen-7B)
50\%
16\%
17\%
18\%
9\%
29\%
ICL (text)
RAG (text)
MEM0 (text)
MEMMACHINE (text)
per-user LoRA (parametric)
\textbf{User as Engram} (parametric)